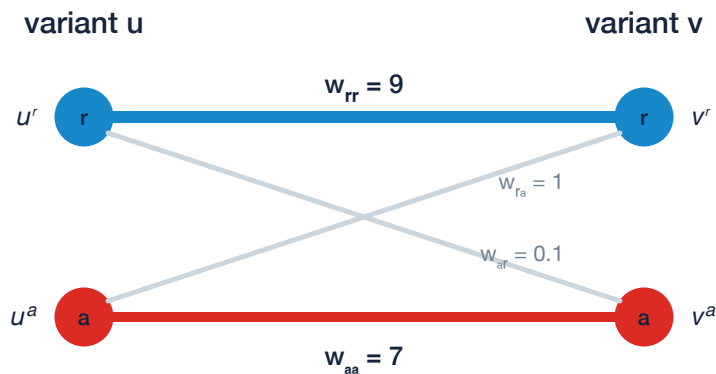


a Four allele-pair weights between two variants



$P = w_{rr} + w_{aa} = 16$ (cis: same haplotype carries r and r)

$Q = w_{ra} + w_{ar} = 1.1$ (trans)

similarity $s = \min(P, Q) / \max(P, Q) = 0.07$

Per read: +1 if both bases $\geq Q12$, else +0.1 (grey edges here carry a low-quality read).

d Exported graph (DOT)

```
1001.1 -> 1543.1 [label=16.0]
```

```
1001.2 -> 1543.2 [label=1.1]
```

position.allele $\rightarrow$ position.allele, label = accumulated haplotype vote weight at the source;
one edge per voting pair; parsed by the GNN module together with INFO/PE, H1, H2.

b Vote rules for the pair (u, v)

$s > 0.7$	no vote (uninformative pair)
$s > 0.3$ for SNV-5mC pairs	no vote
$0.1 < s \leq 0.7$	vote for larger of P, Q; weight 1
$s \leq 0.1$, or one side has no support	vote; weight 20 (near-unanimous)
u is a tandem-repeat indel	vote; weight 0.1

The vote is expressed as a haplotype for v using the haplotype already assigned to u: P favours v^r on $\text{hap}(u^r)$, Q favours v^r on $\text{hap}(u^a)$.

c Single-read guard

Among the votes received by v, count those whose pair is supported by a single read ($P + Q \leq 1$). If more than three such votes exist, h_1 and h_2 are recomputed from high-consistency ($s < 0.2$), non-indel votes only, preventing one chimeric ultra-long read from dominating a sparse region.